claude-windows / 7d6bbb83-42c0-459b-9e23-e14e96d4cb2d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7d6bbb83-42c0-459b-9e23-e14e96d4cb2d\subagents\workflows\wf_1dc2acd6-052\agent-aa21527a82fa27941.jsonl`
- SHA-256: `908a1b3fd13b3fa87f450f8451073501f4039d4cf5e2833bdcdc4f245a702c33`
- Source modified: `2026-06-13T20:25:32+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `wf_1dc2acd6-052`
- session_id: `7d6bbb83-42c0-459b-9e23-e14e96d4cb2d`

Transcript

1. user (2026-06-13T20:21:31.588Z)

You are an ADVERSARIAL fact-checker. Theme: "Constraint/consistency-based correction & post-hoc verification of model outputs ("ML linter"), human-in-the-loop".

Here are a teammate's survey findings (JSON):
{"theme":"Constraint/consistency-based correction & post-hoc verification of model outputs (\"ML linter\"), human-in-the-loop","our_pattern":"A report-first RECONCILER: a post-hoc pass over a persisted rally DECISION (segmenter output) that detects where the decision contradicts the persisted per-frame EVIDENCE (shuttle x,y,visibility/state; person tracks; audio hits; window features), and proposes auditable corrections ? truncate a segment whose shuttle went inactive before its end; drop a segment with zero net-crossings; split an over-merged segment with a long internal dead-time gap; tighten boundary slack to serve-contact. It REPORTS an issues list + a corrected set (never silently rewrites), corrections are A/B-measured for temporal-IoU lift BEFORE auto-apply, idempotent, fixed rule precedence, human-in-the-loop. Cues feeding the upstream scorer are (value,confidence) feature columns weighted by a tiny logistic regression ? \"signal not verdict, no special-casing\" ? and the expensive frame detection is persisted once so re-scoring/reconciling is pure offline (\"persist once, re-score forever\"). Graded with leave-one-video-out (LOVO) vs scarce golden intervals.","key_papers":[{"title":"Model Assertions for Monitoring and Improving ML Models","authors":"Daniel Kang, Deepti Raghavan, Peter Bailis, Matei Zaharia","year":"2020","venue":"MLSys 2020","url_or_id":"arXiv:2003.01668","what_it_does":"Introduces 'model assertions' ? arbitrary user-written functions over a model's input/output that fire when an error is likely, including TEMPORAL CONSISTENCY assertions over video (e.g. an object that rapidly changes class across frames). Used for runtime monitoring (catches HIGH-CONFIDENCE errors that uncertainty methods miss), for active-learning sample selection, and to generate weak labels from violated consistency assertions to retrain. Primarily flags/reports for human review rather than silently rewriting.","overlap":"strong","overlap_explanation":"This is the closest framing match to our reconciler. Our 'detect where the DECISION contradicts persisted EVIDENCE' IS a model assertion over the rally output (shuttle-inactive-before-end, zero-net-crossings, internal-dead-time-gap are consistency assertions). Both are report-first, both exploit temporal/cross-signal consistency, both turn violations into a feedback signal (their weak labels ? our golden-eval lift loop). DIFFERENCE: model assertions stop at flag + (optionally) weak-label-for-retraining; they do NOT propose a structured CORRECTION to the output (truncate/split/drop), don't A/B-measure each correction rule's lift before auto-applying, and have no fixed rule-precedence/idempotency machinery. We should adopt their vocabulary ('model assertion', 'consistency assertion', 'high-confidence error') and cite them as the parent framing."},{"title":"HoloClean: Holistic Data Repairs with Probabilistic Inference","authors":"Theodoros Rekatsinas, Xu Chu, Ihab F. Ilyas, Christopher Ré","year":"2017","venue":"PVLDB Vol.10 / VLDB 2017","url_or_id":"arXiv:1702.00820","what_it_does":"Takes a dirty dataset + integrity constraints (denial constraints) + optional external signals, detects tuples that violate constraints, and produces REPAIRS via a single probabilistic model. Crucially it estimates the MAP value AND the marginal/confidence for each repair, uses low-confidence repairs to solicit additional user feedback in a principled way.","overlap":"partial","overlap_explanation":"Strong conceptual analog to the correction half of our reconciler: constraint-violation DETECTION +

proposed REPAIR + a CONFIDENCE per repair + human-in-the-loop on low-confidence ones. The 'detect contradiction against rules, propose a fix, surface confidence, ask a human when unsure' loop is exactly ours. DIFFERENCES: HoloClean repairs tabular cells, not temporal intervals; it unifies repairs in one global probabilistic-inference model rather than a fixed-precedence rule pipeline that is A/B-gated; and it auto-repairs rather than report-first. Worth adopting: emit a per-correction confidence and route low-confidence corrections to a human queue instead of auto-applying uniformly."},{"title":"Data Validation for Machine Learning","authors":"Eric Breck, Neoklis Polyzotis, Sudip Roy, Steven Whang, Martin Zinkevich","year":"2019","venue":"MLSys 2019 (TensorFlow Data Validation / TFX)","url_or_id":"https://mlsys.org/Conferences/2019/doc/2019/167.pdf","what_it_does":"Production data-validation system (TFDV in TFX): codifies expected properties as a SCHEMA/constraints, compares incoming data statistics to the schema, raises an ANOMALIES report (missing features, out-of-range, type errors, training/serving skew, drift), and proposes schema updates for a human to accept/reject. Report-then-human-fix by design.","overlap":"partial","overlap_explanation":"Matches the 'report-first linter' half: declare constraints, detect violations against persisted statistics, emit a structured anomalies report for a human, and let the system SUGGEST a fix (schema change) that a human ratifies ? directly parallel to our issues-report + proposed-correction + human-in-the-loop. DIFFERENCE: it validates DATA/inputs against a schema, not a model's structured temporal OUTPUT against cross-modal evidence, and it has no temporal-IoU/lift A/B gate. Adopt: their 'schema as version-controlled, human-ratified constraint set' and treating each reconciler rule as a named, owner-acknowledgeable anomaly type."},{"title":"Unit Testing Data with Deequ (Automating Large-Scale Data Quality Verification)","authors":"Sebastian Schelter, Dustin Lange, Philipp Schmidt, Meltem Celikel, Felix Biessmann, Andreas Grafberger","year":"2018","venue":"PVLDB / Amazon (Deequ)","url_or_id":"https://www.amazon.science/publications/unit-testing-data-with-deequ","what_it_does":"Declarative 'unit tests for data': users (or an automated constraint-suggestion profiler) define constraints; the engine verifies them at scale and reports pass/fail with metrics. Includes automated constraint SUGGESTION by profiling data.","overlap":"weak","overlap_explanation":"Supports the 'linter' metaphor and naming directly ? declarative, named constraints over data with a verification report, plus AUTO-SUGGESTED constraints from profiling. Same spirit as our 'register a producer/constraint, get a report'. DIFFERENCE: data-side, tabular, no temporal-interval correction, no lift-measured auto-apply. Useful as terminology precedent ('unit tests for data', 'constraint suggestion') and as a reminder we could auto-mine candidate reconciler rules from the evidence rather than hand-authoring all of them."},{"title":"Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF)","authors":"Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, Hirokatsu Kataoka","year":"2020","venue":"WACV 2021 (arXiv 2020)","url_or_id":"arXiv:2007.06866","what_it_does":"Action Segment Refinement Framework: a boundary-regression branch (BRB) whose predicted action boundaries REFINE a frame-wise action-segmentation branch (ASB), specifically to cut OVER-SEGMENTATION (spurious micro-segments) and sharpen boundaries. Reports segmental EDIT distance and segmental F1 (the metrics this field expects).","overlap":"partial","overlap_explanation":"This is the temporal-segmentation field's canonical answer to the exact failure modes our reconciler targets (over-segmentation, sloppy boundaries) ? our 'split over-merged segment', 'tighten boundary slack to serve-contact' are the boundary-refinement problem. DIFFERENCE: ASRF learns refinement end-to-end inside the model and is NOT a report-first, evidence-cross-checking, A/B-gated post-hoc pass; it has no human-in-the-loop and no auditability. Adopt their FRAMING and especially their METRICS ? segmental F1@{10,25,50} and segmental edit distance are what reviewers in this space expect alongside (or instead of) raw temporal-IoU."},{"title":"Snorkel: Rapid Training Data Creation with Weak Supervision","authors":"Alexander Ratner, Stephen H. Bach, Henry Ehrenberg, Jason Fries, Sen Wu, Christopher Ré","year":"2017","venue":"PVLDB Vol.11 / VLDB 2018","url_or_id":"arXiv:1711.10160","what_it_does":"Users write many noisy 'labeling functions' (heuristics that vote or abstain) with unknown accuracies/correlations; a LABEL MODEL denoises and combines them WITHOUT ground truth into probabilistic labels, learning each source's accuracy rather than hand-weighting.","overlap":"partial","overlap_explanation":"Direct analog of our 'SIGNAL NOT VERDICT / no special-casing' principle: each cue is a noisy voter and a learned model weights them ? Snorkel's label model is the data-programming version of our logistic-regression scorer. DIFFERENCE: Snorkel is about MANUFACTURING training labels from heuristics absent ground truth; we have (scarce) golden labels and supervise directly, and Snorkel models inter-source CORRELATIONS explicitly (we likely don't). Adopt: model cue correlations (correlated cues double-count and miscalibrate the scorer) and consider Snorkel/MeTaL-style label modeling to bootstrap labels from our cheap reconciler rules on UNlabeled matches."},{"title":"Confident Learning: Estimating Uncertainty in Dataset Labels","authors":"Curtis G. Northcutt, Lu Jiang, Isaac L. Chuang","year":"2021","venue":"JAIR (Journal of AI Research)","url_or_id":"arXiv:1911.00068","what_it_does":"Model-agnostic method that uses a model's out-of-sample predicted probabilities to estimate the joint distribution of noisy vs true labels, IDENTIFIES likely label errors (prune-by-noise-rate), ranks them, and proposes corrected candidates ? the basis of cleanlab.","overlap":"weak","overlap_explanation":"Same 'use the model's own evidence to flag where the recorded label/decision is probably wrong, rank by margin, propose a candidate' loop ? applicable to auditing our scarce GOLDEN intervals themselves (vendor labels are noisy) and to ranking reconciler-flagged segments by how strongly evidence contradicts the decision. DIFFERENCE: it targets classification label noise, not temporal-interval boundaries or cross-modal consistency. Adopt: rank reconciler issues by a confident-learning-style margin so humans review the most contradicted segments first."},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","authors":"Yi-Hua Hsu, Chih-Chang Yu, Hsu-Yung Cheng","year":"2024","venue":"Sensors (MDPI), 24(13):4372","url_or_id":"https://doi.org/10.3390/s24134372 (PMC11244353)","what_it_does":"Fuses TrackNet shuttle-trajectory hit detection with a YOLOv7 swing-action detector via a Shot Refinement Algorithm (SRA): case-based rules CROSS-CHECK trajectory hits against action detections (e.g. 'only one hit can exist within a swing', drop a trajectory hit outside any swing action, keep the one nearest the peak-confidence action frame) to correct false-positive/false-negative hits.","overlap":"partial","overlap_explanation":"The domain-closest work and a real analog of our reconciler's CORE idea ? cross-check one signal's output against another signal's evidence and CORRECT contradictions. Our 'shuttle inactive before segment end' / 'zero net-crossings' are the rally-level cousins of their shot-level 'hit outside any swing action'. DIFFERENCES (important and in our favor): (1) it operates at SHOT level, not rally start/end / dead-time-vs-in-play segmentation; (2) corrections are HAND-CODED case-rules in the decision path ? exactly the 'special-casing' we avoid; (3) NOT report-first, NO A/B lift measurement before applying, no learned weighting, no human-in-the-loop, no idempotency/precedence formalism. So our contribution vs this paper is precisely the methodology layer (signal-not-verdict + report-first + measured-before-apply), not the cross-check idea itself."}],"established_terms":["model assertions / consistency assertions (Kang et al., MLSys'20) ? the standard name for 'a function that fires when model output violates an expected property'","high-confidence errors ? errors a confident model makes that uncertainty/confidence thresholds miss (the value-add of assertion/consistency checks)","data repair / error detection-and-repair, integrity constraints, denial constraints (HoloClean / DB cleaning) ? the framing for 'detect violation, propose repair'","data validation / schema anomalies / training-serving skew (TFDV, Polyzotis et al.) ? production 'report-then-human-ratify' constraint checking","unit tests for data / declarative data-quality constraints / constraint suggestion (Deequ, Great Expectations) ? the 'linter for data' metaphor and auto-mined constraints","action segment refinement / over-segmentation / boundary regression; segmental F1@{10,25,50} and segmental edit distance (ASRF, MS-TCN lineage) ? the temporal-segmentation field's standard refinement vocabulary and METRICS","data programming / label model / labeling functions (Snorkel) ? learned denoising of many noisy heuristic voters (your 'signal not verdict')","confident learning / prune-by-noise-rate (Northcutt et al., cleanlab) ? model-evidence-based detection of likely-wrong labels/decisions","leave-one-group-out / leave-one-subject-out (LOSO/LOGO) cross-validation ? the canonical name for your LOVO; 'group-wise CV' is the literature term"],"techniques_to_adopt":[{"technique":"Explicitly brand the reconciler as 'model assertions / consistency assertions over the rally decision' and cite Kang et al. ? frame each rule (shuttle-inactive-before-end, zero-net-crossings, internal-dead-time-gap) as a named consistency assertion.","why":"Gives the design an established, defensible name and lineage instead of bespoke 'reconciler' terminology; reviewers immediately understand it, and it correctly positions you as EXTENDING model assertions (you add structured correction + lift-gated auto-apply, which they lack).","from_paper":"Model Assertions for Monitoring and Improving ML Models (Kang et al., 2020)","effort":"low"},{"technique":"Report segmental F1@{10,25,50} and segmental edit distance alongside temporal-IoU, and benchmark the reconciler against a learned boundary-refinement baseline (ASRF-style).","why":"Raw temporal-IoU alone under-credits the over-segmentation/boundary fixes your reconciler targets; segmental F1 and edit distance are exactly what the temporal-action-segmentation field uses to quantify over-segmentation and boundary quality, making your lift numbers comparable and credible.","from_paper":"Alleviating Over-segmentation Errors by Detecting Action Boundaries / ASRF (Ishikawa et al., 2020)","effort":"medium"},{"technique":"Attach a per-correction CONFIDENCE/margin to every reconciler proposal and route only high-confidence ones to auto-apply, sending low-confidence ones to a human queue ranked by margin.","why":"HoloClean and confident learning both show that surfacing repair confidence and reviewing the most-contradicted cases first is the principled way to mix auto-repair with human-in-the-loop; uniform auto-apply on a rule is riskier than confidence-gated apply, especially on a 6-match corpus.","from_paper":"HoloClean (Rekatsinas et al., 2017) + Confident Learning (Northcutt et al., 2021)","effort":"medium"},{"technique":"Treat the constraint/assertion set as a version-controlled, owner-ratified 'schema' of named anomaly types, with an automated CONSTRAINT-SUGGESTION step that profiles persisted evidence to propose candidate reconciler rules.","why":"TFDV and Deequ show production value in (a) making constraints first-class, versioned, human-acknowledged artifacts and (b) auto-mining candidate constraints from data so you don't hand-author every rule ? directly applicable to growing your reconciler rule set beyond the few you wrote by hand.","from_paper":"Data Validation for ML / TFDV (Polyzotis et al., 2019) + Unit Testing Data with Deequ (Schelter et al., 2018)","effort":"medium"},{"technique":"Model cue CORRELATIONS in the scorer (and consider a Snorkel/MeTaL-style label model to bootstrap labels on unlabeled matches from your reconciler rules).","why":"Your cues (shuttle-state, net-crossings, person two-sidedness) are correlated; an unregularized logistic regression on a tiny set will double-count correlated cues and miscalibrate. Snorkel's label model is the established way to denoise correlated heuristic voters, and it can manufacture weak rally labels on the 5 unlabeled-of-6 matches to enlarge your training signal.","from_paper":"Snorkel (Ratner et al., 2017)","effort":"high"},{"technique":"Use violated assertions as WEAK LABELS to improve the upstream scorer, not just to correct outputs.","why":"Kang et al. show consistency-assertion violations can be turned into weak supervision that retrains the model for up to ~46% quality gain ? closing your loop so the reconciler feeds the scorer instead of only patching its outputs.","from_paper":"Model Assertions (Kang et al., 2020)","effort":"medium"}],"pitfalls":["Small held-out + learned weights overfit: the late-fusion literature explicitly warns that learning modality/cue weights by regression on a small held-out set is UNSTABLE and overfits. With ~6 matches and a logistic regression over many correlated cues, your learned weights (and any per-cue calibration) can be dominated by one match. Mitigate with strong L2/L1 regularization, far fewer effective features than groups, and reporting weight variance across LOVO folds.","LOVO is necessary but not sufficient ? distributional bias still biases leave-one-out: recent work shows leave-one-out CV suffers distributional bias (the held-out group shifts the train-set label distribution), inflating or deflating estimates, worse under class imbalance (rallies vs dead-time is imbalanced). Report variance/CIs across the 6 folds, not a single point, and watch for the imbalance interaction.","Correlated cues double-count: if net-crossing count, two-sidedness, and shuttle-in-air all co-fire on the same rallies, a naive scorer over-weights that bundle and a reconciler built on the same cues won't catch the resulting errors (no independent evidence). Snorkel's lesson: model source correlations or your 'independent' evidence isn't independent.","Hand-coded correction rules creep back into the decision path (the badminton SRA failure mode): the closest domain work (Hsu et al.) implements correction as five hand-coded case rules ? exactly the special-casing you forbid. The risk is that 'report-first reconciler rules' silently become production if/else branches with fixed thresholds that don't generalize across venues/cameras. Keep them flag-gated, lift-measured, and learn thresholds where possible.","'Report-first' can decay into silent auto-rewrite: TFDV/HoloClean keep a human ratification step precisely because auto-repair on confident-but-wrong cases is dangerous (HoloClean's own precision ~90% means ~10% bad repairs). Without per-rule lift gating AND a confidence gate AND idempotency tests, an over-eager reconciler rule can systematically truncate correct long rallies. Measure lift per rule on golden data before any auto-apply, and keep rules idempotent with fixed precedence so order doesn't change outcomes.","Refinement that optimizes the wrong metric: ASRF shows boundary refinement can raise segmental F1/edit while frame accuracy barely moves (and vice versa). If you only watch temporal-IoU, a reconciler that fixes over-segmentation may show no IoU lift yet be a real win ? and conversely an IoU-chasing rule can shred segment structure. Track both interval and segmental metrics so you don't reward the wrong behavior.","Assertion/repair set as unowned tech debt: Deequ/TFDV treat constraints as versioned, owned artifacts. Ad-hoc reconciler rules accumulate, conflict, and rot; without a registry, precedence spec, and per-rule provenance/lift record, the 'linter' becomes an unauditable pile of thresholds."],"verdict":"novel-combination","verdict_rationale":"novel-combination","search_queries_used":["model assertions Kang validating machine learning predictions runtime monitoring","constraint-based correction structured prediction enforcing consistency neural network outputs","Snorkel weak supervision label model denoising labeling functions data programming","data validation constraints machine learning pipelines TFX schema anomaly detection Polyzotis","temporal action segmentation post-processing refinement over-segmentation boundary smoothing","human-in-the-loop verification ML predictions proposals verify against evidence active learning correction sports video","HoloClean error detection repair data cleaning integrity constraints probabilistic inference","badminton rally segmentation shuttlecock trajectory TrackNet temporal localization stroke detection","learned weighting of noisy heuristic rules vs hand-coded rules calibration late fusion pitfalls multi-signal","grouped cross-validation leave-one-subject-out data leakage small dataset weak supervision evaluation pitfalls","\"unit tests for data\" Deequ Great Expectations declarative constraints data quality automated suggestion","consistency-based label cleaning confident learning detecting label errors pruning Northcutt"]}

Your job is to be skeptical and protect against two failure modes: (1) FABRICATED citations, (2) OVERCLAIMED novelty.

DO, using WebSearch/WebFetch (load via ToolSearch "select:WebSearch,WebFetch" if needed):
1. For each cited paper, search its EXACT title (+authors) to confirm it is a REAL paper. Mark verified_real
   yes/no/uncertain and flag any that don't resolve in fabrication_flags. Be strict: a plausible-sounding title
   that you cannot find is a fabrication flag.
2. For every "novel-combination" or "genuine-gap" verdict, actively try to REFUTE it: run fresh searches for the
   CLOSEST prior art that would downgrade the verdict toward "reinventing"/"partial-overlap". Default to assuming
   prior art exists until your searches come up empty. List any closer_prior_art you find (with real ids).
3. Give a corrected_verdict + rationale. Add extra_course_corrections if your searches surfaced adopt/avoid notes.

Return the structured object.

2. assistant (2026-06-13T20:21:33.720Z)

I'll fact-check this survey. Let me start by loading the web search tools.

3. assistant (2026-06-13T20:21:38.979Z)

I'll fact-check each citation by searching for its exact title, then attempt to refute the "novel-combination" verdict. Let me start with parallel searches for the cited papers.

4. user (2026-06-13T20:21:49.704Z)

Web search results for query: "Model Assertions for Monitoring and Improving ML Models Kang Raghavan Bailis Zaharia MLSys 2020"

Links: [{"title":"Google Scholar","url":"https://scholar.google.com/scholar?q=Daniel+Kang,+Deepti+Raghavan,+Peter+Bailis,+and+Matei+Zaharia.+2020.+Model+assertions+for+monitoring+and+improving+ML+models.+Proceedings+of+Machine+Learning+and+Systems+2+(2020),+481--496."},{"title":"Model Assertions for Monitoring and Improving ML Models","url":"https://arxiv.org/pdf/2003.01668"},{"title":"[PDF] Model Assertions for Monitoring and Improving ML Models | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Model-Assertions-for-Monitoring-and-Improving-ML-Kang-Raghavan/1924b2473d3cb866f88a988eb745e828b30b6294"},{"title":"Model Assertions for Monitoring and Improving ML Models - ADS","url":"https://ui.adsabs.harvard.edu/abs/2020arXiv200301668K/abstract"},{"title":"Model Assertions for Monitoring and Improving ML Models","url":"https://proceedings.mlsys.org/paper_files/paper/2020/hash/e851ca7b43815718fbbac8afb2246bf8-Abstract.html"},{"title":"Model Assertions for Monitoring and Improving ML Model","url":"https://www.researchgate.net/publication/339675163_Model_Assertions_for_Monitoring_and_Improving_ML_Model"},{"title":"[2003.01668] Model Assertions for Monitoring and Improving ML Models","url":"https://arxiv.org/abs/2003.01668"},{"title":"Model Assertions for Monitoring and Improving ML Models","url":"https://mlsys.org/media/Slides/mlsys/2020/balla(03-16-30)-03-17-45-1437-mlperf_training.pdf"},{"title":"Book","url":"https://proceedings.mlsys.org/paper/2020"},{"title":"Model Assertions for Monitoring and Improving ML Models","url":"https://people.eecs.berkeley.edu/~matei/papers/2020/mlsys_model_assertions.pdf"}]

Great! I found the paper you're looking for. Here's information about it:

Model Assertions for Monitoring and Improving ML Models

This paper was authored by Daniel Kang, Deepti Raghavan, Peter Bailis, and Matei Zaharia and published in the MLSys 2020 proceedings.

Overview

The paper proposes model assertions, a new abstraction that adapts the classical use of program assertions as a way to monitor and improve ML models. Model assertions are arbitrary functions over a model's input and output that indicate when errors may be occurring, for example, a function that triggers if an object rapidly changes its class in a video.

Key Contributions

The authors propose methods of using model assertions at all stages of ML system deployment, including runtime monitoring, validating labels, and continuously improving ML models. For runtime monitoring, they show that model assertions can find high confidence errors, where a model returns the wrong output with high confidence, which uncertainty-based monitoring techniques would not detect.

For training, they propose a bandit-based active learning algorithm that can sample from data flagged by assertions and show that it can reduce labeling costs by up to 40% over traditional uncertainty-based methods. They also propose an API for generating "consistency assertions" and weak labels for inputs where the consistency assertions fail, and show that these weak labels can improve relative model quality by up to 46%.

The evaluation was performed on four real-world tasks with video, LIDAR, and ECG data.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-13T20:21:50.254Z)

Web search results for query: "HoloClean Holistic Data Repairs with Probabilistic Inference Rekatsinas Chu Ilyas Ré 2017"

Links: [{"title":"HoloClean: Holistic Data Repairs with Probabilistic Inference","url":"http://www.vldb.org/pvldb/vol10/p1190-rekatsinas.pdf"},{"title":"Figure 6 from HoloClean: Holistic Data Repairs with Probabilistic Inference | Semantic Scholar","url":"https://www.semanticscholar.org/paper/HoloClean%3A-Holistic-Data-Repairs-with-Probabilistic-Rekatsinas-Chu/698410acff560a5eb9e880e951a96c75e48ac9bf/figure/9"},{"title":"HoloClean: Holistic Data Repairs with Probabilistic Inference","url":"https://arxiv.org/pdf/1702.00820"},{"title":"[PDF] HoloClean: Holistic Data Repairs with Probabilistic Inference | Semantic Scholar","url":"https://www.semanticscholar.org/paper/HoloClean:-Holistic-Data-Repairs-with-Probabilistic-Rekatsinas-Chu/464195f162b6755959084869395234 94d0efb2ab"},{"title":"[1702.00820] HoloClean: Holistic Data Repairs with Probabilistic Inference","url":"https://arxiv.org/abs/1702.00820"},{"title":"HoloClean: holistic data repairs with probabilistic inference: Proceedings of the VLDB Endowment: Vol 10, No 11","url":"https://dl.acm.org/doi/10.14778/3137628.3137631"},{"title":"HoloClean: Holistic Data Repairs with Probabilistic Inference | Request PDF","url":"https://www.researchgate.net/publication/313366531_HoloClean_Holistic_Data_Repairs_with_Probabilistic_Inference"},{"title":"dblp: HoloClean: Holistic Data Repairs with Probabilistic Inference.","url":"https://dblp.org/rec/journals/corr/RekatsinasCIR17.html"},{"title":"Analyzing Deviations from Monotonic Trends through Database Repair","url":"https://arxiv.org/pdf/2512.08526"},{"title":"A Theoretical Framework for AI-driven data quality monitoring in   high-volume data environments","url":"https://arxiv.org/pdf/2410.08576"}]

I found the paper you're looking for! Here are the search results:

HoloClean: Holistic Data Repairs with Probabilistic Inference

HoloClean is a framework for holistic data repairing driven by probabilistic inference, authored by Theodoros Rekatsinas and Christopher Ré (Stanford University) and Xu Chu and Ihab F. Ilyas (University of Waterloo).

Key Contributions:

HoloClean unifies qualitative data repairing, which relies on integrity constraints or external data sources, with quantitative data repairing methods, which leverage statistical properties of the input data. Given an inconsistent dataset as input, HoloClean automatically generates a probabilistic program that performs data repairing.

Scalability and Performance:

The framework introduces a series of optimizations which ensure that inference over HoloClean's probabilistic model scales to instances with millions of tuples. HoloClean finds data repairs with an average precision of ~90% and an average recall of above ~76% across diverse datasets, yielding an average F1 improvement of more than 2× against state-of-the-art methods.

Publication:

The paper was published in the Proceedings of the VLDB Endowment, Vol. 10, No. 11 in 2017. The full paper is available at several sources including the VLDB official publication and arXiv.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-13T20:21:53.108Z)

Both verified. Note HoloClean precision ~90% matches the survey's pitfall claim. Let me check the next batch.

7. user (2026-06-13T20:22:01.805Z)

Web search results for query: "Unit Testing Data with Deequ Schelter Lange Schmidt Celikel Biessmann Grafberger Amazon"

Links: [{"title":"Unit Testing Data with Deequ","url":"https://bibbase.org/network/publication/schelter-biessmann-lange-rukat-schmidt-seufert-brunelle-taptunov-unittestingdatawithdeequ-2019"},{"title":"[PDF] Unit Testing Data with Deequ | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Unit-Testing-Data-with-Deequ-Schelter-Biessmann/ffba151794ea1ed7d44229d9f926824fc37e9c4f"},{"title":"Unit Testing Data with Deequ | Proceedings of the 2019 International Conference on Management of Data","url":"https://dl.acm.org/doi/abs/10.1145/3299869.3320210"},{"title":"GitHub - awslabs/deequ: Deequ is a library built on top of Apache Spark for defining \"unit tests for data\", which measure data quality in large datasets. · GitHub","url":"https://github.com/awslabs/deequ"},{"title":"dblp: Unit Testing Data with Deequ.","url":"https://dblp.org/rec/conf/sigmod/SchelterBLRSSBT19.html"},{"title":"Unit testing data with Deequ - Amazon Science","url":"https://www.amazon.science/publications/unit-testing-data-with-deequ"},{"title":"Unit Testing Data with Deequ","url":"https://assets.amazon.science/12/62/7a066bf44199afea06ec73f23ad0/unit-testing-data-with-deequ.pdf"},{"title":"Unit Testing Data with Deequ | Sebastian Schelter","url":"https://ssc.io/publication/unit-testing-data-with-deequ-sigmod/"},{"title":"Auto-Validate by-History: Auto-Program Data Quality Constraints to Validate Recurring Data Pipelines","url":"https://arxiv.org/pdf/2306.02421"},{"title":"Unit Testing Data with Deequ | Request PDF","url":"https://www.researchgate.net/publication/333858706_Unit_Testing_Data_with_Deequ"}]

I've found comprehensive information about "Unit Testing Data with Deequ." Here's what the search results show:

Publication Details

The paper "Unit Testing Data with Deequ" was authored by Schelter, S., Biessmann, F., Lange, D., Rukat, T., Schmidt, P., Seufert, S., Brunelle, P., & Taptunov, A., and was published in the Proceedings of the 2019 International Conference on Management of Data, pages 1993?1996, 2019 by ACM.

What is Deequ?

Deequ is an Apache Spark-based library for automating the verification of data quality at scale.

The library provides a declarative API, which combines common quality constraints with user-defined validation code, and thereby enables "unit tests for data".

Key Features

Deequ meets the requirements of production use cases at Amazon, and scales to datasets with billions of records if the constraints to evaluate are chosen carefully. Deequ is a library built on top of Apache Spark for defining "unit tests for data", which measure data quality in large datasets, and Python users may also be interested in PyDeequ, a Python interface for Deequ.

The research represents work by researchers at Amazon that addresses the critical challenge of verifying data quality at scale in modern data-driven organizations.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-13T20:22:02.796Z)

Web search results for query: "Data Validation for Machine Learning Breck Polyzotis Roy Whang Zinkevich MLSys 2019 TFDV"

Links: [{"title":"Data Validation for Machine Learning","url":"https://proceedings.mlsys.org/paper_files/paper/2019/hash/928f1160e52192e3e0017fb63ab65391-Abstract.html"},{"title":"Breck, E., Polyzotis, N., Roy, S., Whang, S.E. and Zinkevich, M. (2019) Data Validation for Machine Learning. - References - Scientific Research Publishing","url":"https://www.scirp.org/(S(vtj3fa45qm1ean45vvffcz55))/reference/referencespapers.aspx?referenceid=3292273"},{"title":"Data Validation for Machine Learning | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Data-Validation-for-Machine-Learning-Breck-Polyzotis/2d57b38ff6fa651cc932a29f0c7ca72e9f7a8cf9"},{"title":"Data Validation for Machine Learning","url":"https://research.google/pubs/pub47967/"},{"title":"REVOLUTIONIZING DEEP LEARNING WITH SYSML: The Power of Data Validation in Machine Learning | by Ansh Mittal | Medium","url":"https://medium.com/@anshm18111996/revolutionizing-deep-learning-with-sysml-the-power-of-data-validation-in-machine-learning-10299f25daf0"},{"title":"Towards Interactively Improving ML Data Preparation Code via \"Shadow Pipelines\"","url":"https://arxiv.org/pdf/2404.19591"},{"title":"CM-Explorer: Dissecting Data Ingestion Problems Niels Bylois UHasselt","url":"https://www.vldb.org/pvldb/vol16/p3958-bylois.pdf"},{"title":"dblp: Eric Breck","url":"https://dblp.org/pid/16/2435.html"},{"title":"\"We Have No Idea How Models will Behave in Production until Production\":   How Engineers Operationalize Machine Learning","url":"https://arxiv.org/pdf/2403.16795"},{"title":"Adaptive Machine Learning Frameworks for Data Quality Monitoring: From Anomaly Detection to Continuous Pipeline Validation | International Journal of Research and Applied Innovations","url":"https://ijrai.org/index.php/ijrai/article/view/481"}]

I found the research paper you're looking for. Here are the key details:

"Data Validation for Machine Learning" (MLSys 2019)

The paper is authored by Eric Breck, Neoklis Polyzotis, Sudip Roy, Steven Whang, and Martin Zinkevich, published at MLSys in 2019.

Overview

While machine learning research has focused on improving training and inference algorithms, there is less attention to monitoring data quality fed to machine learning, even though errors in input data can nullify benefits on speed and accuracy. The paper presents a data validation system designed to detect anomalies in data fed into machine learning pipelines.

Implementation (TFDV)

The paper introduces a data validation system deployed at Google that is an integral part of

TFX, an end-to-end ML platform. The system was designed to detect anomalies in data and was later used by 100s of product teams to monitor and validate several petabytes of production data daily.

The paper is available through the MLSys proceedings and has become influential in the field, with several subsequent works building upon its ideas for systematic data validation in machine learning.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-13T20:22:08.672Z)

Both verified. Note: the "Unit Testing Data with Deequ" paper venue is SIGMOD 2019 (not "PVLDB / Amazon" as the survey states ? there is a separate VLDB "Automating Large-Scale Data Quality Verification" paper). The author list in the survey is slightly off (survey lists "Meltem Celikel, Felix Biessmann, Andreas Grafberger" but actual is Biessmann, Lange, Rukat, Schmidt, Seufert, Brunelle, Taptunov). This is a minor discrepancy ? the paper is real and the survey conflates two related Deequ papers. Let me continue checking the remaining citations.

10. user (2026-06-13T20:22:16.215Z)

Web search results for query: "Alleviating Over-segmentation Errors by Detecting Action Boundaries ASRF Ishikawa Kasai Aoki Kataoka WACV 2021"

Links: [{"title":"Alleviating Over-segmentation Errors by Detecting Action
Boundaries","url":"https://yiskw713.github.io/asrf/"},{"title":"Alleviating over-segmentation
errors by detecting action boundaries ? Keio
University","url":"https://keio.elsevierpure.com/en/publications/alleviating-over-segmentation-
errors-by-detecting-action-boundari-2"},{"title":"WACV 2021 Open Access Repository","url":"https
://openaccess.thecvf.com/content/WACV2021/html/Ishikawa_Alleviating_Over-
Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.html"},{"title":"GitHub -
yiskw713/asrf: Code for ''Alleviating Over-segmentation Errors by Detecting Action Boundaries''
accepted in WACV2021","url":"https://github.com/yiskw713/asrf"},{"title":"[PDF] Alleviating
Over-segmentation Errors by Detecting Action Boundaries | Semantic
Scholar","url":"https://www.semanticscholar.org/paper/Alleviating-Over-segmentation-Errors-by-
Detecting-Ishikawa-Kasai/4fed0235614df46fa0c53e05b71e9939b53a1bdf"},{"title":"Alleviating Over-
segmentation Errors by Detecting Action Boundaries","url":"https://openaccess.thecvf.com/content
/WACV2021/papers/Ishikawa_Alleviating_Over-
Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.pdf"},{"title":"Alleviating
Over-Segmentation Errors by Detecting Action
Boundaries","url":"https://papertalk.org/papertalks/27315"},{"title":"Alleviating Over-
segmentation Errors by Detecting Action
Boundaries","url":"https://arxiv.org/pdf/2007.06866"},{"title":"Alleviating Over-segmentation
Errors by Detecting Action Boundaries | Request
PDF","url":"https://www.researchgate.net/publication/352386005_Alleviating_Over-
segmentation_Errors_by_Detecting_Action_Boundaries"}]

I found the research paper you're looking for. Here's what the search revealed:

Paper Summary

The paper "Alleviating Over-segmentation Errors by Detecting Action Boundaries" was authored by Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka and was published in WACV 2021.

Main Contributions

The paper proposes an Action Segment Refinement Framework (ASRF) that consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB). The ASB classifies video frames with action classes, while the BRB regresses the action boundary probabilities.

The model outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score.

Resources

- The official project page is available at: https://yiskw713.github.io/asrf/
- The code is publicly available on GitHub: https://github.com/yiskw713/asrf
- The full paper is available in the IEEE/CVF Open Access Repository and on arXiv

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-13T20:22:16.684Z)

Web search results for query: "Snorkel Rapid Training Data Creation with Weak Supervision Ratner Bach Ehrenberg Fries Wu Ré VLDB"

Links: [{"title":"Snorkel: Rapid Training Data Creation With Weak Supervision | Page 10 of 10 | Snorkel AI","url":"https://snorkel.ai/research-paper/snorkel-rapid-training-data-creation-with-weak-supervision-2/"},{"title":"[PDF] Snorkel: Rapid Training Data Creation with Weak Supervision | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Snorkel:-Rapid-Training-Data-Creation-with-Weak-Ratner-Bach/18bc1d4271abe8dd6e16179cdb06524a4f396e16"},{"title":"Snorkel: Rapid Training Data Creation with Weak Supervision Alexander Ratner","url":"https://cs.brown.edu/people/sbach/files/ratner-vldb17.pdf"},{"title":"Snorkel: rapid training data creation with weak supervision: Proceedings of the VLDB Endowment: Vol 11, No 3","url":"https://dl.acm.org/doi/10.14778/3157794.3157797"},{"title":"Snorkel: rapid training data creation with weak supervision - PubMed","url":"https://pubmed.ncbi.nlm.nih.gov/32214778/"},{"title":"[1711.10160] Snorkel: Rapid Training Data Creation with Weak Supervision","url":"https://arxiv.org/abs/1711.10160"},{"title":"Snorkel: Rapid Training Data Creation with Weak Supervision","url":"https://arxiv.org/pdf/1711.10160"},{"title":"Learning Hyper Label Model for Programmatic Weak Supervision","url":"https://arxiv.org/pdf/2207.13545"},{"title":"Snorkel: Rapid Training Data Creation with Weak Supervision | Stanford DAWN","url":"https://dawnd9.sites.stanford.edu/publications/snorkel/snorkel-rapid-training-data-creation-weak-supervision"},{"title":"An Extensible Multimodal Multi-task Object Dataset with Materials","url":"https://arxiv.org/pdf/2305.14352"}]

I've found the paper you're looking for. Here's what I found:

"Snorkel: Rapid Training Data Creation with Weak Supervision" is a VLDB 2017 paper by Alexander Ratner, Stephen H Bach, Henry Ehrenberg, Jason Fries, Sen Wu, and Christopher Ré.

The paper addresses the fact that labeling training data is increasingly the largest bottleneck in deploying machine learning systems, and presents Snorkel, a system that enables users to train state-of-the-art models without hand labeling any training data.

Instead of manual labeling, users write labeling functions that express arbitrary heuristics with unknown accuracies and correlations, and Snorkel denoises their outputs without access to ground truth by incorporating the first end-to-end implementation of the data programming machine learning paradigm.

The paper was published in the Proceedings of the VLDB Endowment, Volume 11, Issue 3, and is available through multiple sources including arXiv, ACM Digital Library, and the Snorkel AI website.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-13T20:22:26.136Z)

Both verified (ASRF arXiv:2007.06866, Snorkel arXiv:1711.10160, PVLDB Vol.11 No.3 ? all match). Now the last two cited papers, including the domain-closest one.

13. user (2026-06-13T20:22:33.502Z)

Web search results for query: "Enhancing Badminton Game Analysis Shot Refinement Fusion Shuttlecock Tracking Hit Detection Hsu Yu Cheng Sensors 2024"

Links: [{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.mdpi.com/1424-8220/24/13/4372"},{"title":"Fine reconstruction of badminton swing dynamic trajectory assisted by event camera | Scientific Reports","url":"https://www.nature.com/articles/s41598-026-46443-8"},{"title":"Future Prediction of Shuttlecock Trajectory in Badminton Using Player's Information - PubMed","url":"https://pubmed.ncbi.nlm.nih.gov/37233318/"},{"title":"A Model for Analyzing Shuttlecock Landing Position Using Real-Time Object Detection Techniques | Journal of Applied Science and Emerging Technology","url":"https://ph01.tci-thaijo.org/index.php/JASCI/article/view/265289"},{"title":"(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"},{"title":"MATCHMOTION: AN AI-POWERED MOBILE APPLICATION FOR AUTOMATED BADMINTON VIDEO ANALYSIS USING YOLOV8 OBJECT DETECTION","url":"https://www.researchgate.net/publication/404597971_MATCHMOTION_AN_AI-POWERED_MOBILE_APPLICATION_FOR_AUTOMATED_BADMINTON_VIDEO_ANALYSIS_USING_YOLOV8_OBJECT_DETECTION"},{"title":"Game-Based Training Model: Does It Improve Fundamental Badminton Young Athletes' Skills? | Journal of Coaching and Sports Science","url":"https://journal.foundae.com/index.php/jcss/article/view/905"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"}]

I found the research paper you were looking for. Here are the details:

Title: Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera

Authors: Yi-Hua Hsu, Chih-Chang Yu, and Hsu-Yung Cheng

Publication: Sensors, 24(13), 4372 (2024). https://doi.org/10.3390/s24134372

Research Summary:

The study focuses on extracting the flight trajectory of the shuttlecock in a single turn in badminton games as important for automated sports analytics, proposing a novel method to extract shots in badminton games from a monocular camera.

Methodology: The method uses TrackNet, a deep neural network designed for tracking small objects, to extract the flight trajectory of the shuttlecock; the YOLOv7 model to identify whether the player is swinging; and proposes a shot refinement algorithm to obtain the correct hitting moment, accounting for detection misses and false detections from both systems.

Results: The proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches with 1582 rallies of Badminton World Federation (BWF) match videos. This represents a significant improvement compared to using TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-13T20:22:37.111Z)

Web search results for query: "Confident Learning Estimating Uncertainty in Dataset Labels Northcutt Jiang Chuang JAIR cleanlab"

Links: [{"title":"[1911.00068v2] Confident Learning: Estimating Uncertainty in Dataset Labels","url":"https://arxiv.org/abs/1911.00068v2"},{"title":"[PDF] Confident Learning: Estimating Uncertainty in Dataset Labels | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Confident-Learning:-Estimating-Uncertainty-in-Northcutt-Jiang/cbaaa1154c491f9da2f050d3c22970e15bb7b52b"},{"title":"[1911.00068] Confident Learning: Estimating Uncertainty in Dataset Labels","url":"https://arxiv.org/abs/1911.00068"},{"title":"Confident Learning: Estimating Uncertainty in Dataset Labels | Request PDF","url":"https://www.researchgate.net/publication/350909446_Confident_Learning_Estimating_Uncertainty_in_Dataset_Labels"},{"title":"Confident Learning: Estimating Uncertainty in Dataset Labels","url":"https://dl.acm.org/doi/pdf/10.1613/jair.1.12125"},{"title":"Automated Cleanup of the ImageNet Dataset by Model Consensus,   Explainability and Confident Learning","url":"https://arxiv.org/pdf/2103.16324"},{"title":"[1911.00068] Confident Learning: Estimating Uncertainty in Dataset Labels","url":"https://ar5iv.labs.arxiv.org/html/1911.00068"},{"title":"Con?dent Learning: Estimating Uncertainty in Dataset Labels Curtis G. Northcutt","url":"https://arxiv.org/pdf/1911.00068"}]

I found detailed information about the paper you're searching for. Here are the key results:

Paper Details

The paper "Confident Learning: Estimating Uncertainty in Dataset Labels" is authored by Curtis G. Northcutt, Lu Jiang, and Isaac L. Chuang.

Key Concepts

Confident learning (CL) is an approach which focuses on label quality by characterizing and identifying label errors in datasets, based on the principles of pruning noisy data, counting with probabilistic thresholds to estimate noise, and ranking examples to train with confidence.

The work generalizes CL by building on the assumption of a classification noise process, to directly estimate the joint distribution between noisy (given) labels and uncorrupted (unknown) labels.

Implementation and Results

The generalized CL, open-sourced as cleanlab, is provably consistent across reasonable conditions, and experimentally performant on ImageNet and CIFAR, outperforming seven recent approaches when label noise is non-uniform.

CL was employed on ImageNet to quantify ontological class overlap (e.g., estimating 645 "missile" images are mislabeled as their parent class "projectile"), and moderately increase model accuracy (e.g., for ResNet) by cleaning data prior to training.

The paper appears to have been published in the Journal of Artificial Intelligence Research (JAIR), based on the search results referencing it as a published work.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-06-13T20:22:41.281Z)

All 9 cited papers are verified real, with correct authors, venues, and IDs. The badminton paper's "Shot Refinement Algorithm (SRA)" claim is confirmed. Now the critical adversarial part: I'll try to refute the "novel-combination" verdict by searching for closer prior art ? specifically work that combines report-first/post-hoc verification + structured temporal-interval correction + measured-before-apply + human-in-the-loop, and any sports/video work that post-hoc corrects rally/segment boundaries against persisted multi-signal evidence.

16. user (2026-06-13T20:22:49.767Z)

Web search results for query: "post-hoc correction temporal segmentation boundaries against multi-modal evidence human-in-the-loop report proposed corrections measured lift before apply"

Links: [{"title":"[2510.09945] Explainable Human-in-the-Loop Segmentation via Critic Feedback Signals","url":"https://arxiv.org/abs/2510.09945"},{"title":"Temporal Action Segmentation: An Analysis of Modern Techniques","url":"https://arxiv.org/pdf/2210.10352"},{"title":"Temporal Uncertainty Localization to Enable Human-in-the-loop Analysis of Dynamic Contrast-enhanced Cardiac MRI Datasets - PubMed","url":"https://pubmed.ncbi.nlm.nih.gov/37664410/"},{"title":"MedUHIP: Towards Human-In-the-Loop Medical Segmentation","url":"https://arxiv.org/html/2408.01620v1"},{"title":"[2408.01620] MedUHIP: Towards Human-In-the-Loop Medical Segmentation","url":"https://arxiv.org/abs/2408.01620"},{"title":"Temporal Uncertainty Localization to Enable Human-in-the-loop Analysis of Dynamic Contrast-enhanced Cardiac MRI Datasets","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10473819/"},{"title":"Boundary-Centric Active Learning for Temporal Action Segmentation","url":"https://arxiv.org/html/2604.15173v1"}]

Based on the search results, I found several relevant papers on human-in-the-loop segmentation and temporal analysis:

Key Findings:

1. **Human-in-the-Loop Interactive Segmentation**: One approach enables interventional learning through targeted human corrections of segmentation outputs, treating these corrections as signals that help the model learn more robust features.

2. **Temporal Action Segmentation**: Several post-processing approaches target improving action boundaries, including BCN which features a pooling operator to smooth noisy boundary predictions, and ASRF which refines boundaries with a complementary network branch that directly regresses boundary locations.

3. **Dynamic Quality Control for Temporal Data**: A space-time uncertainty metric has been proposed as a quality control tool, where the top 10% most uncertain segmentations are referred to human experts for refinement, resulting in significant improvements in accuracy.

4. **Multi-Modal Medical Segmentation**: An uncertainty-aware model integrated with human-in-the-loop interaction proposes several plausible segmentations, while human interaction iteratively modifies the segmentation under clinician supervision.

5. **Boundary-Centric Active Learning**: A hierarchical approach ranks and queries unlabeled videos using predictive uncertainty, then detects candidate transitions and selects top boundaries via a novel boundary score that fuses neighborhood uncertainty, class ambiguity, and temporal predictive dynamics.

The search results show active research in human-in-the-loop correction frameworks, particularly for temporal segmentation and boundary detection with multi-modal inputs.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. user (2026-06-13T20:22:50.508Z)

Web search results for query: "rally segmentation refinement post-processing consistency constraints shuttle trajectory dead-time gap audit corrections sports video"

Links: [{"title":"From Broadcast to Minimap: Achieving State-of-the-Art SoccerNet Game   State Reconstruction","url":"https://arxiv.org/pdf/2504.06357"},{"title":"EventAnchor: Reducing Human Interactions in Event Annotation of Racket   Sports Videos","url":"https://arxiv.org/pdf/2101.04954"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular

Camera","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"},{"title":"Towards Athlete Fatigue Assessment from Association Football Videos","url":"https://arxiv.org/pdf/2604.05636"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"Correction: Yin et al. Improvement of SAM2 Algorithm Based on Kalman Filtering for Long-Term Video Object Segmentation. Sensors 2025, 25, 4199","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12390045/"},{"title":"TT3D: Table Tennis 3D Reconstruction","url":"https://arxiv.org/pdf/2504.10035"},{"title":"SmartTennisTV: Automatic indexing of tennis videos","url":"https://arxiv.org/pdf/1801.01430"},{"title":"Video Segmentation in Computer Vision: A Short Guide [+Examples]","url":"https://www.lightly.ai/blog/video-segmentation"},{"title":"How I Built an Automated Pickleball Rally Splitter Using Python and YOLOv8 | by Vinod Polinati | Medium","url":"https://medium.com/@vinodpolinati/how-i-built-an-automated-pickleball-rally-splitter-using-python-and-yolov8-b890a44650a4"}]

I've completed the web search for your query. The results focus on sports video analysis, particularly rally segmentation and trajectory reconstruction techniques. Here are the key findings:

Rally Segmentation:
Rally segmentation in racket sports videos can be performed by detecting timestamps of score changes, achieving 98.5% accuracy.

Post-Processing & Refinement:
Post-processing plays a critical role in refining tracking results by addressing tracklet fragmentation, identity swaps, and temporal inconsistencies, with the objective to concatenate short tracklets into longer coherent trajectories while ensuring consistency across attributes.

Trajectory Gaps & Dead-Time Handling:
Gaps in player trajectories are addressed using linear interpolation to fill missing positions when players temporarily leave the camera's field of view or are occluded.

Shuttle/Ball Trajectory Reconstruction:
Methods leverage detected court, shuttlecock track, and player poses to segment video frames into shots and reconstruct 3D trajectory for each shot.

Sports Video Analysis:
Post-processing techniques include short-gap interpolation that linearly interpolates missing positions for short temporal gaps, and refinement techniques like erosion, dilation, and Gaussian blurring remove rough edges and fill small gaps.

The search results reveal a comprehensive framework for analyzing sports videos with emphasis on consistency constraints and gap handling in trajectory data.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. assistant (2026-06-13T20:22:55.173Z)

Interesting ? found two candidate closer-prior-art items worth probing: "EventAnchor: Reducing Human Interactions in Event Annotation of Racket Sports Videos" (domain-exact, human-in-the-loop correction of racket-sports events) and "Boundary-Centric Active Learning for Temporal Action Segmentation." Let me examine EventAnchor closely and search for the verification/assertion-over-structured-output combination more directly.

19. user (2026-06-13T20:23:04.186Z)

I cannot reliably analyze this document. The provided content appears to be a PDF file encoded in binary/compressed format (indicated by the `%PDF-1.5` header and FlateDecode streams), which renders as unintelligible text rather than readable content.

To properly answer your questions about EventAnchor's approach to:
- Detecting contradictions between model decisions and per-frame evidence
- Proposing auditable corrections for human review

- Reporting issues versus silent rewrites
- Measuring correction lift
- Handling human-in-the-loop annotation correction

I would need either:
1. A plaintext or properly decoded version of the paper
2. A PDF file that can be properly parsed
3. The paper's abstract or methods section in readable format

If you can provide the document in a readable format, I'd be happy to analyze EventAnchor's actual methodology around these specific aspects.

[Binary content (application/pdf, 3.2MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7d6bbb83-42c0-459b-9e23-e14e96d4cb2d\tool-results\webfetch-1781382179007-2fedwz.pdf]

20. user (2026-06-13T20:23:06.670Z)

Web search results for query: "verification proposed corrections structured prediction model output against constraints report-first never silently rewrite idempotent rule precedence A/B measured lift before auto-apply ML linter"

Links: [{"title":"Eidoku: A Neuro-Symbolic Verification Gate for LLM Reasoning via Structural Constraint Satisfaction","url":"https://arxiv.org/pdf/2512.20664"},{"title":"Learning Constraints for Structured Prediction Using ...","url":"https://aclanthology.org/2020.acl-main.438.pdf"},{"title":"Formal verification of tree-based machine learning models for lateral spreading","url":"https://arxiv.org/pdf/2603.16983"},{"title":"Pushing the Boundaries of Natural Reasoning: Interleaved Bonus from Formal-Logic Verification","url":"https://arxiv.org/pdf/2601.22642"},{"title":"Structured prediction","url":"https://en.wikipedia.org/wiki/Structured_prediction"},{"title":"CCTU: A Benchmark for Tool Use under Complex Constraints","url":"https://arxiv.org/pdf/2603.15309"},{"title":"Generalization Analysis on Learning with a Concurrent Verifier","url":"https://arxiv.org/pdf/2210.05331"},{"title":"Constraint Reasoning Embedded Structured Prediction","url":"https://jmlr.org/papers/volume23/21-1484/21-1484.pdf"},{"title":"mllint ? Linter for Machine Learning projects","url":"https://bvobart.github.io/mllint/"}]

Based on the search results, I found relevant information about verification of structured prediction models against constraints. Here's what the research shows:

Verification and Constraint Checking in Structured Prediction

Verification should ask whether a statement can be embedded into the contextual semantic structure without violating its constraints, rather than relying solely on statistical confidence measures.

Hard constraints prevent violations during training by restricting the hypothesis space, but formal verification audits a trained model by checking whether its learned function satisfies specified properties, returning either a proof or a concrete counterexample.

Verify-Fix-Verify Workflow

Research describes a verify-fix-verify loop where an unconstrained model is trained for accuracy, physical specifications are formally verified, counterexamples are analyzed to determine needed constraints or corrections, and targeted fixes are applied.

Constraint Issues and Corrections

During data verification, identified issues can be categorized into conflicts among constraints, mismatches between constraints and scenario structures, and conflicts between constraints and tools, with manual corrections needed for each category.

ML Linting

ML linters measure a project's adherence to ML best practices, such as those from SE4ML and Google's Rules for ML, though your specific query about "report-first never silently rewrite" and "A/B measured lift before auto-apply" appears to reference proprietary or internal tooling not covered in the available academic literature.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. assistant (2026-06-13T20:23:11.873Z)

The PDF fetch failed (binary). Let me get EventAnchor's content from a readable source (ar5iv or the HTML abstract) and also probe one more angle ? the "Auto-Validate by-History" paper appeared earlier, which auto-programs data-quality constraints; and let me check the "Concurrent Verifier" framing which is conceptually close to measured constraint-application.

22. user (2026-06-13T20:23:14.945Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://ar5iv.org/abs/2101.04954
Redirect URL: https://ar5iv.labs.arxiv.org/abs/2101.04954
Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://ar5iv.labs.arxiv.org/abs/2101.04954"
- prompt: "Summarize EventAnchor's method. Specifically: does it cross-check a model's event/segment decisions against persisted per-frame evidence (ball trajectory, player position) to find contradictions, propose corrections, report an issues list, and route them to a human? Is it report-first vs silent-rewrite? Does it measure correction lift (e.g. IoU) before applying? Is it about an interactive annotation tool or an automated reconciler?"

23. user (2026-06-13T20:23:24.989Z)

Web search results for query: ""persist once" re-score offline persisted per-frame detections cached features rescore segmenter decision against evidence badminton rally reconciler"

Links: [{"title":"A high-precision and efficient method for badminton action detection in sports using You Only Look Once with Hourglass Network - ScienceDirect","url":"https://www.sciencedirect.com/science/article/abs/pii/S0952197624013356"},{"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"},{"title":"Towards Real-Time Analysis of Broadcast Badminton Videos","url":"https://arxiv.org/pdf/2308.12199"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/pdf/2307.16000"},{"title":"Automated service scene detection for Badminton game analysis Using CHLAC and MRA | Request PDF","url":"https://www.researchgate.net/publication/290557755_Automated_service_scene_detection_for_Badminton_game_analysis_Using_CHLAC_and_MRA"},{"title":"(PDF) Improving Badminton Action Recognition Using Spatio-Temporal Analysis and a Weighted Ensemble Learning Model","url":"https://www.researchgate.net/publication/385709017_Improving_Badminton_Action_Recognition_Using_Spatio-Temporal_Analysis_and_a_Weighted_Ensemble_Learning_Model"},{"title":"(PDF) Making the World?s Fastest Racket Sport even Better: A Systematic Review of Artificial Intelligence-based Objective Player Performance Assessment in Badminton","url":"https://www.researchgate.net/publication/402667183_Making_the_World's_Fastest_Racket_Sport_even_Better_A_Systematic_Review_of_Artificial_Intelligence-based_Objective_Player_Performance_Assessment_in_Badminton"}]

The search results returned several research papers about badminton video analysis, but they don't appear to contain the specific technical terminology from your query. The results focus on badminton action detection, rally detection, hit-frame detection, and stroke recognition, but don't mention the specific concepts like "persist once," "re-score offline," "cached features rescore," "segmenter decision," or "rally reconciler" that appear in your search query.

The papers discuss various aspects of badminton analysis including:
- Rally detection (distinguishing gameplay from replays)
- Hit-frame and shot detection
- Stroke recognition and classification
- Feature extraction methods

However, the exact combination of technical terms in your query doesn't appear in these results. This could indicate that:
1. These terms may be from a very specialized or newer research area not yet widely indexed
2. They might be from a proprietary system or internal documentation
3. The specific terminology combination might be from academic work that hasn't been widely published yet

Would you like me to search for a different variation of these terms, or do you have additional context about what system or research project this relates to?

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

24. assistant (2026-06-13T20:23:31.577Z)

The "persist once, re-score forever" + reconciler combination did not surface any closer prior art in the badminton domain. Let me fetch EventAnchor via the redirect to confirm whether it's a closer match, and run one final refutation search targeting the most defensible novelty claim (the methodology layer: report-first + measured-before-apply structured correction).

25. user (2026-06-13T20:23:41.957Z)

EventAnchor's Approach: Interactive Annotation, Not Automated Reconciliation

EventAnchor is **an interactive annotation and visual-analytics tool**, not an automated reconciler with measured-before-apply validation.

Method Overview

EventAnchor operates at three levels:

1. **Object-level**: Computer vision extracts ball position, player pose, court location
2. **Event-level**: Algorithms detect key moments (ball-racket contact, ball-court contact) and create "anchors"
3. **Context-level**: Humans manually add domain knowledge through interactive tools

Key Distinction: Report-First with User Verification

The system generates anchor suggestions but requires human calibration. As the paper states: "Anchors are automatically detected by algorithms and inevitably contain errors. The calibration box is used to calibrate the time of an anchor."

Users can:
- Adjust timestamps via the calibration box
- Delete erroneous anchors
- Add missing anchors manually

No Cross-Check for Contradictions

EventAnchor does **not** systematically cross-check model decisions against persisted frame-level evidence, propose structured corrections, or generate issues lists. Instead, it surfaces suggestions and trusts user judgment to accept, modify, or reject them.

Verification Approach

The system measures user task performance *after* annotation (via experiments), not correction lift *before* applying changes. One participant noted concern: users might accept algorithmic suggestions without verification.

Bottom line: Human-in-the-loop interactive tool emphasizing efficiency gains, not automated validation.

26. user (2026-06-13T20:23:44.430Z)

Web search results for query: "model assertions extended with automatic repair structured output correction measured before apply human-in-the-loop video segmentation novel combination"

Links: [{"title":"Automatic High-quality Verilog Assertion Generation through   Subtask-Focused Fine-Tuned LLMs and Iterative Prompting","url":"https://arxiv.org/pdf/2411.15442"},{"title":"Automated Video Segmentation Machine Learning Pipeline","url":"https://arxiv.org/html/2507.07242v1"},{"title":"Segmentation Rectification for Video Cutout via One-Class Structured Learning","url":"https://arxiv.org/pdf/1602.04906"},{"title":"Automatic video image segmentation","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8422775"},{"title":"Live Interactive Training for Video Segmentation","url":"https://arxiv.org/html/2603.26929v1"},{"title":"Automated Video Segmentation Machine Learning Pipeline | Proceedings of the Digital Production Symposium","url":"https://dl.acm.org/doi/10.1145/3744199.3744635"},{"title":"Video Segmentation in Computer Vision: A Short Guide [+Examples]","url":"https://www.lightly.ai/blog/video-segmentation"}]

Based on my search results, I found several relevant papers and resources related to your query about combining model assertions, automatic repair, structured output correction, and human-in-the-loop video segmentation:

Key Findings:

1. **Automated Video Segmentation with Structured Output**: An automated video segmentation pipeline creates temporally consistent instance masks and is deployed using containerization and leveraging a structured output format. The structured nature of the output enables template work and streamlines downstream processes.

2. **Automatic Repair and Iterative Correction**: An iterative repair method has been incorporated where the RTL code and assertions are provided to an LLM to correct semantic errors, assertions are submitted to a compiler for verification, and if incorrect, compilation errors are sent back to the LLM for correction until a correct assertion is obtained.

3. **Human-in-the-Loop Interactive Video Segmentation**: A Live Interactive Training (LIT) framework enables models to learn online from human corrections at inference time, with LIT-LoRA implementing this by continually updating a lightweight LoRA module that is rapidly trained on user feedback.

These components represent a novel combination of model verification, automatic correction mechanisms, and interactive human feedback for improving video segmentation systems.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

27. assistant (2026-06-13T20:23:50.895Z)

I now have enough to render a verdict. EventAnchor is confirmed to be an interactive annotation/visual-analytics tool ? NOT an automated reconciler that cross-checks decisions against persisted evidence with measured-before-apply correction. It is closer prior art on the domain + human-in-the-loop axis, but it does not collapse the survey's specific combination. No prior art surfaced that bundles: (report-first structured temporal-interval correction over a persisted decision) + (cross-modal evidence contradiction detection) + (per-rule A/B lift-gated auto-apply) + (idempotent fixed-precedence) + (learned-cue scorer). The "novel-combination" verdict holds, though I'd note EventAnchor and the badminton SRA paper bound the domain novelty more tightly than the survey credits.

Let me record my findings.

28. user (2026-06-13T20:24:42.804Z)

Output does not match required schema: root: must NOT have additional properties

29. assistant (2026-06-13T20:24:46.458Z)

I included an extra key. Let me resubmit with only the schema-valid properties.

30. user (2026-06-13T20:25:23.699Z)

Structured output provided successfully

31. assistant (2026-06-13T20:25:32.320Z)

Fact-check complete. Summary of findings:

Fabrication check: all 9 cited papers are REAL and verified (correct authors, venues, years, arXiv/DOI IDs). No fabrication flags. One citation-hygiene issue worth fixing: the Deequ entry conflates two real Deequ papers ? "Unit Testing Data with Deequ" (SIGMOD 2019) vs "Automating Large-Scale Data Quality Verification" (PVLDB 2018) ? and mis-assigns the author list/venue between them.

Novelty check: verdict UPHELD as novel-combination, with caveats. Refutation searches found no prior art bundling the full claimed stack (report-first structured temporal-interval correction over a persisted decision + cross-modal evidence contradiction detection + per-rule A/B lift-gating before auto-apply + idempotent fixed-precedence + learned cue scorer). But the survey overstates the *domain* novelty: I surfaced **EventAnchor (arXiv:2101.04954)** ? a racket-sports human-in-the-loop event-correction system the survey missed ? plus the already-cited Hsu et al. cross-check-and-correct work. Together they show the cross-check insight and the human-in-the-loop piece are both prior art in this exact domain, so the defensible contribution is narrowed to the methodology layer (measured-before-apply + signal-not-verdict + auditable rule registry), not the cross-check idea.

Full structured object returned via StructuredOutput.